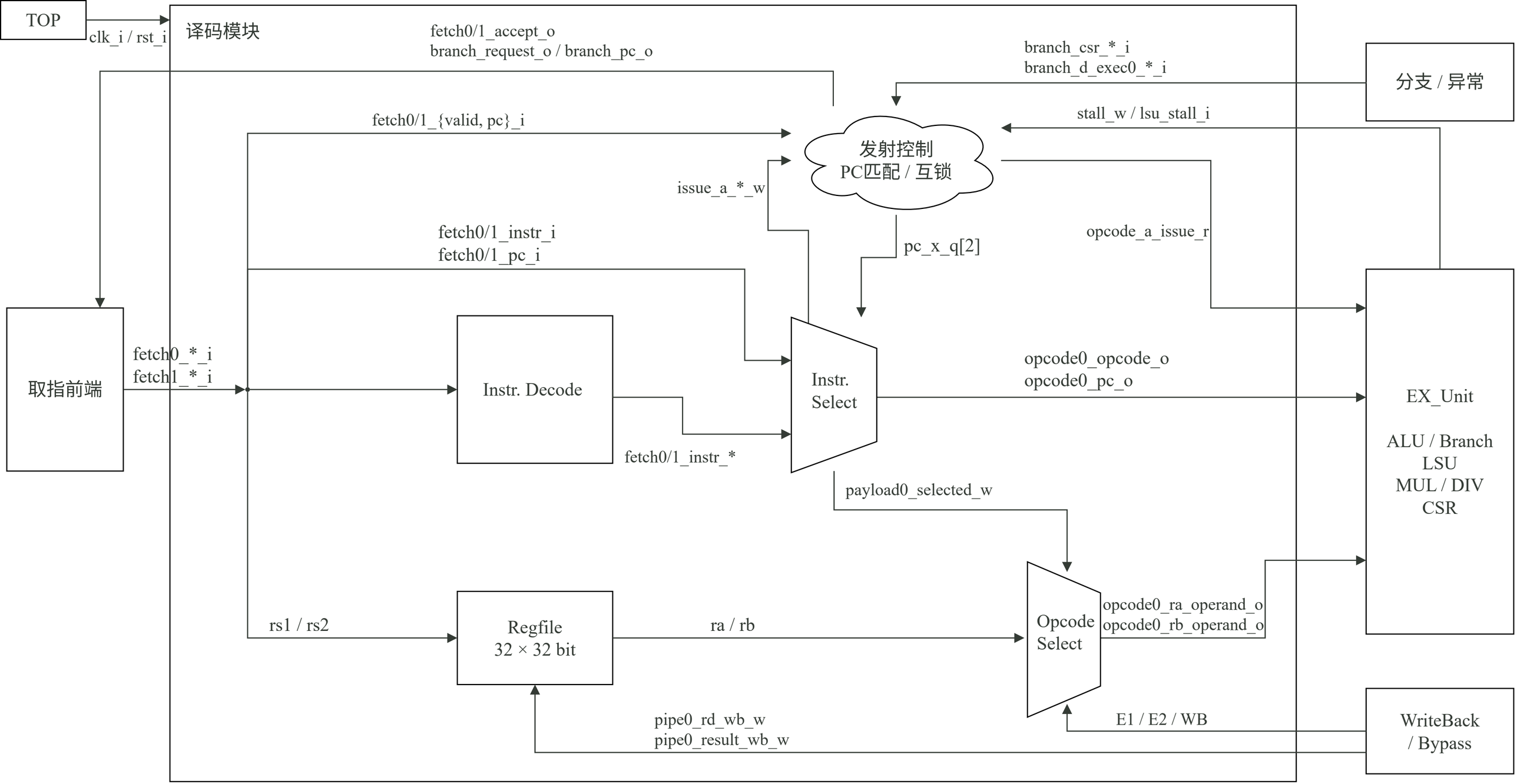

TOP
clk_i / rst_i
译码模块
fetch0/1_accept_o
branch_request_o / branch_pc_o
branch_csr_*_i
branch_d_exec0_*_i
分支 / 异常
fetch0/1_{valid, pc}_i
stall_w / lsu_stall_i
发射控制
PC匹配 / 互锁
issue_a_*_w
fetch0/1_instr_i
fetch0/1_pc_i
pc_x_q[2]
opcode_a_issue_r
取指前端
fetch0_*_i
fetch1_*_i
Instr. Decode
Instr.
Select
opcode0_opcode_o
opcode0_pc_o
EX_Unit
ALU / Branch
LSU
MUL / DIV
CSR
fetch0/1_instr_*
payload0_selected_w
rs1 / rs2
Regfile
32 × 32 bit
ra / rb
Opcode
Select
opcode0_ra_operand_o
opcode0_rb_operand_o
E1 / E2 / WB
WriteBack
/ Bypass
pipe0_rd_wb_w
pipe0_result_wb_w